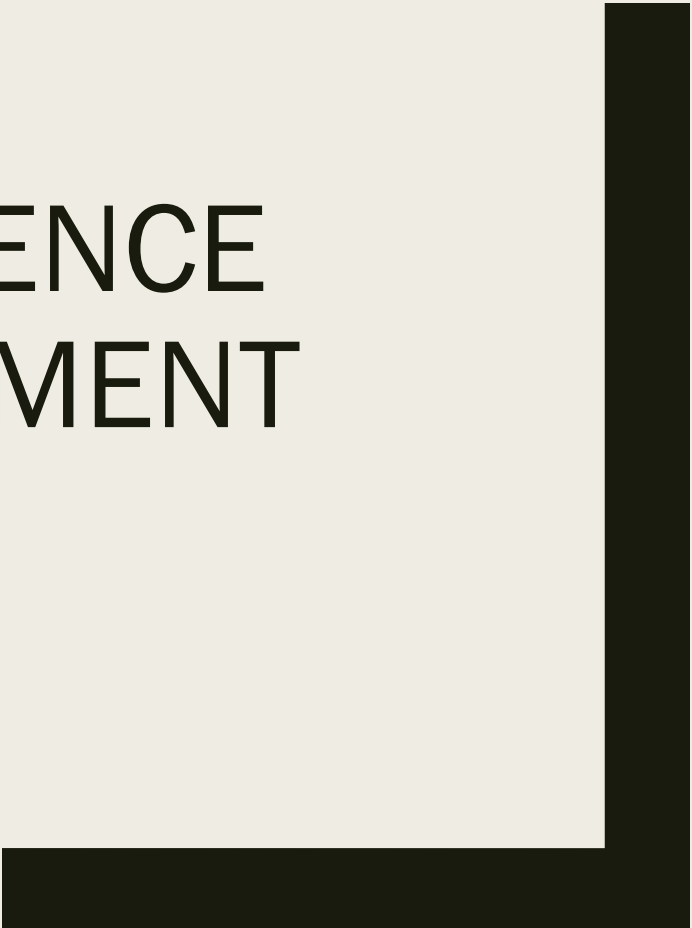




ARTIFICIAL INTELLIGENCE SOFTWARE DEVELOPMENT

CST8510 Week 12
Dr. Hari M Koduvely



Agenda for Today

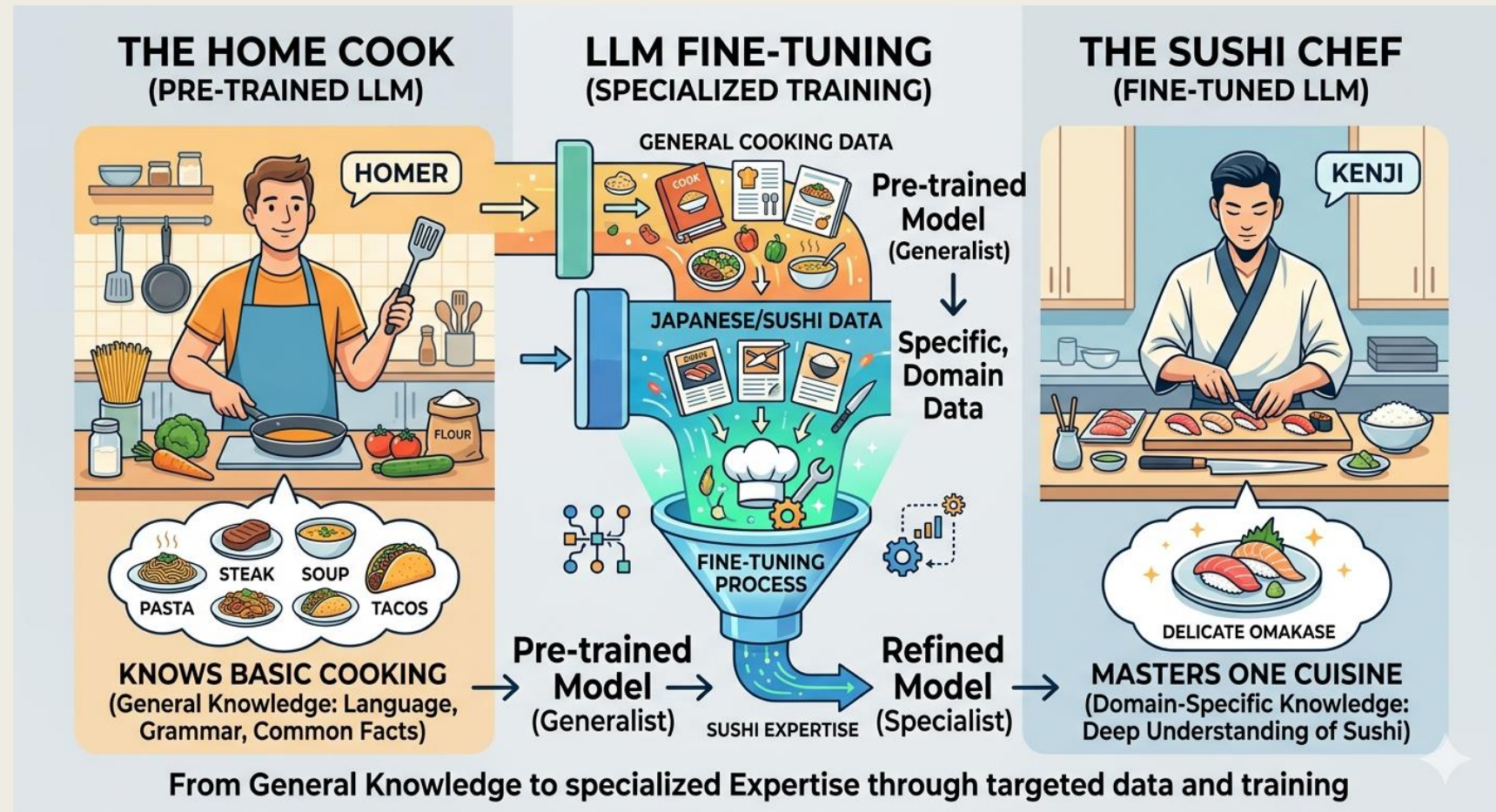
- ❑ Theory: 5:30PM – 7:30PM
 - LLM Fine-Tuning
- ❑ Lab: 7:30PM – 9:30PM
 - Standup Meetings



LLM Fine-Tuning

- Process of tuning a broad general-purpose LLM to perform exceptionally well on specific domain or task.
- Foundation models or Base models are pre-trained on a vast amount of data to perform well on a general tasks.
- However, they would fail miserably when asked about very domain specific questions.
- Example Home Cook Vs Sushi Chef

LLM Fine-Tuning



When to Fine-Tune LLMs?

- LLM needs to perform Complex Tasks
 - E.g. Code like an expert software developer
- Cost and Latency need to be optimized
 - Achieve the same performance using a smaller and more efficient model.
- Critical applications where accuracy and reliability are paramount.
- Adjust Tone, Style & Behaviour to your customer audience
- Enforce Structure and Format
- Domain Adaptation

Do these before Fine-Tuning

- Prompt Engineering
 - ☐ Few shot learning examples
 - ☐ Chain-of-Thought Prompting
 - ☐ Self-consistent and Threshold based prompting
 - ☐ Prompt optimization using OPRO
- Advanced Retrieval Augmented Generation (RAG)
 - ☐ Semantic Chunking
 - ☐ Query Transformation
 - ☐ Fusion Retrieval
 - ☐ Re-ranking of the retrieved context

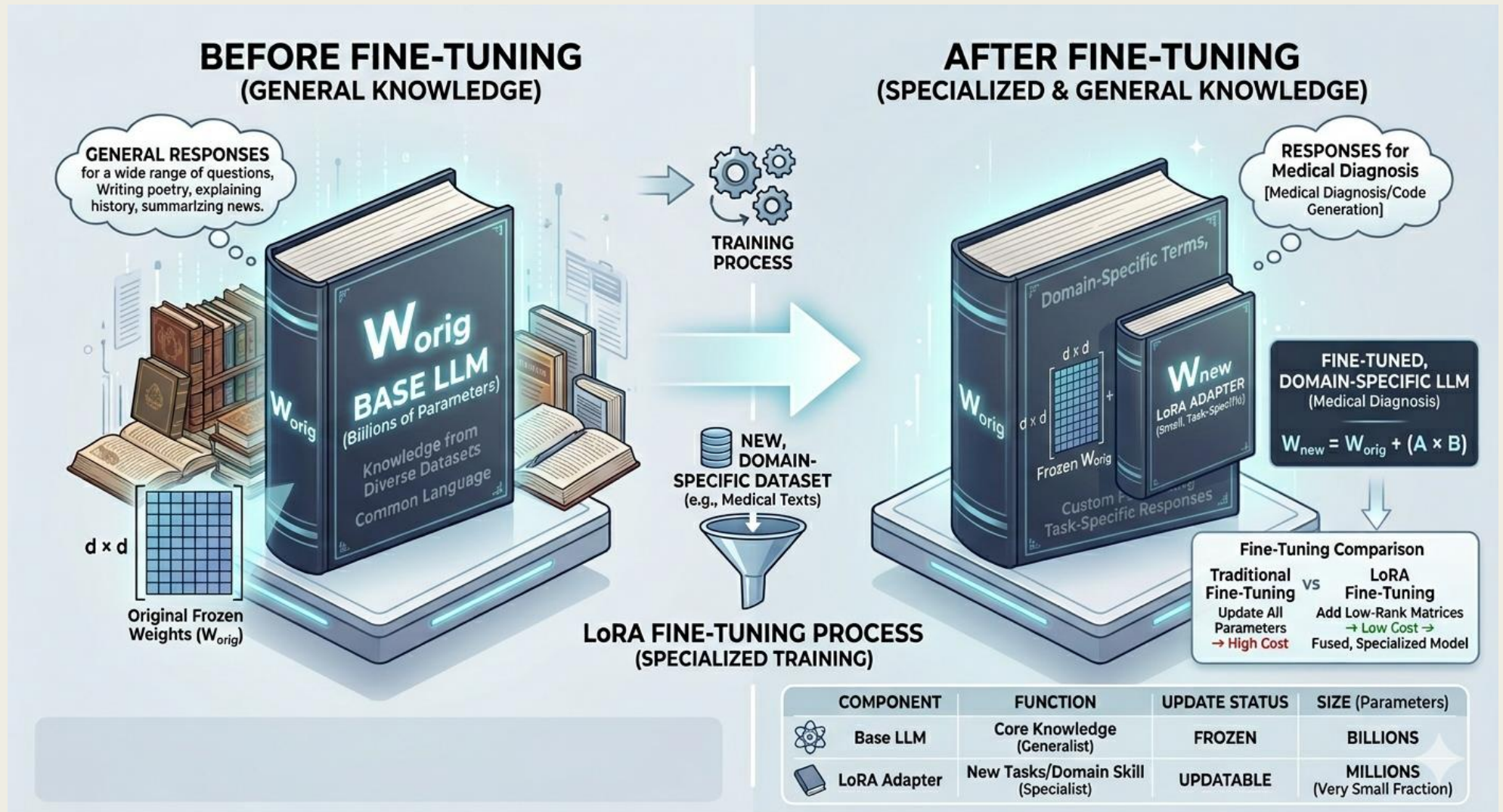
When Not to Do Fine-Tuning

- Data changes too frequently
 - E. g. Stock
- Simple tasks
- Low-quality data
- Privacy Constraints

Different Types of Fine-Tuning

- Full Fine-Tuning
 - ☐ Update all the parameters of an LLM.
 - ☐ Powerful, but expensive
 - ☐ Could result in catastrophic forgetting
- LoRA (Low-Rank Adaptation)
 - ☐ Freezes the original model weights
 - ☐ Weight in selected layers are updated using small, trainable "adapter" modules.
 - ☐ Only the adapters are trained, which is more memory and computationally efficient process.
 - ☐ Adapters can be switched easily to adapt to different domains/tasks.

Low Rank Adaptation (LoRA)



Different Fine-Tuning Frameworks

Feature	Unsloth	Axolotl	LLaMA-Factory	HF PEFT
Primary Goal	Raw Speed / Low VRAM	Reproducibility / Scaling	Ease of Use / All-in-one	Core Logic / Integration
Interface	Python / No-code Studio	YAML Config	Web UI (LlamaBoard)	Python API
Best Hardware	Single GPU (Consumer)	Multi-GPU (H100/A100 clusters)	Flexible	Any
Difficulty	Low (Studio) to Med	High	Low	Med-High
Multi-GPU?	Yes (Pro/Enterprise)	Native / Best	Yes	Yes

Workout Example Using Colab

[Fine-Tuning Llama Model with Cybersecurity Domain Data](#)

Summary of Today's Learning

- ❑ What is Fine-Tuning LLMs.
- ❑ When to and not to fine-tune.
- ❑ LoRA and PEFT
- ❑ Colab implementation of Llama fine-tuning using Cybersecurity Data

Thank You